Locating the Source of Hallucination in LLaVA-1.5: A Three-Track Empirical Investigation of the ViT Visual Encoder

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Abstract. Vision-language models (VLMs) hallucinate objects that are not present in the input image. Most prior work treats this as a decoder-side failure and leaves the visual encoder unexamined. We ask the opposite question: is the ViT encoder already producing degraded representations on images where LLaVA-1.5-7B later hallucinates? We test this through three controlled experiments on 500 COCO 2017 validation images. Track A (attention rollout entropy) finds no relationship between [CLS] attention spread and hallucination outcome ($\rho = 0.003$, $p = 0.94$, Cohen's $d = 0.039$). Track B (linear probes on patch embeddings at four ViT layers) finds the opposite: object discriminability is uniformly lower on hallucinating images at every probed layer ($\Delta\text{acc} \in [-0.039, -0.027]$, Wilcoxon $p < 0.005$ at all layers). Track C is reported in two configurations. The first uses 4-bit NF4 LLaVA-1.5-7B with hidden-state steering at LLM layers 14–15 and stratifies images by Track A entropy quartiles, finding directional VEGAS backfire on diffuse-attention images ($\Delta\text{CHAIR}_i = +0.021$, $P(\Delta \geq 0) = 0.929$). The second uses full fp16 LLaVA-1.5-7B with rollout-derived concentration steering and stratifies by Track B quality groups, finding zero effect on all groups due to near-uniform rollout concentration. Together the three tracks support a refined hypothesis: the ViT encoder fails on hallucinating images in feature content, not in attention geometry, and attention-based steering methods such as VEGAS have a structural blind spot in this failure mode.

1 Introduction

VLMs such as LLaVA-1.5 routinely generate object descriptions that are not present in the image they are conditioned on. Standard benchmarks for hallucination, including POPE [**?**] and CHAIR [**?**], evaluate the textual output and treat the visual encoder as a fixed, reliable upstream component. Methods designed to reduce hallucination, including VEGAS [**?**], build directly on this assumption: VEGAS injects ViT [CLS]-to-patch attention into LLM middle layers as a trusted spatial signal that tells the language model which visual tokens are worth attending to.

This project inverts the assumption. If the ViT is already producing degraded representations on the images that lead to hallucination, then any mitigation method that trusts ViT attention is operating on a signal that cannot distinguish good encodings from bad ones. We call this a structural blind spot: a failure mode that the method cannot detect by construction. The contribution of this paper

is to look inside the ViT on the same 500-image sample that produces hallucinations and check whether the encoder is in fact failing on those images.

Hypothesis. On images where LLaVA-1.5-7B produces a hallucinated object, the ViT encoder's representations, either its attention maps or its patch embeddings, are already measurably degraded relative to clean images, before any token is generated. The hypothesis decomposes into three falsifiable sub-claims, one per track:

1. **Track A.** Hallucinating images have more diffuse [CLS] attention, measured as higher rollout entropy.
2. **Track B.** Hallucinating images have less object-discriminative patch embeddings, measured as lower linear-probe accuracy at intermediate ViT layers.
3. **Track C.** An attention-based steering method (VEGAS) fails specifically on the images Track B identifies as feature-degraded.

The three tracks are designed so that the result is interpretable regardless of which sub-claims hold. A null on Track A combined with a positive on Track B would tell us that the encoder fails in feature content but not in attention geometry. A backfire on Track C would tell us that attention-based mitigation is unsafe in the failure regime. As we will show, the data lands close to this scenario.

Why this is a useful question. Most published hallucination-mitigation methods (VEGAS [**?**], OPERA, VCD, VISTA) intervene at the decoder or inject signals back into the language model. They share an implicit assumption that the visual encoder is reliable. Seo et al. [**?**] found a correlation between ViT token uncertainty and hallucination using adversarial perturbations but did not test a mitigation method against the failure they identified. Our project tests that link end-to-end on a real VLM. Even a partial null is publishable: it relocates the encoder-failure signal from attention to features and identifies a class of methods that cannot see it.

2 Related Work

Hallucination in VLMs. Object hallucination in image captioning was first characterized by CHAIR [**?**], which scores per-image hallucination as the fraction of mentioned object types absent from COCO ground truth. POPE [**?**] probes binary object existence with yes/no questions on three adversarial splits (random, popular, adversarial). We use CHAIR_i as the primary metric throughout because we want a continuous score per image, not a binary probe outcome.

Attention rollout and ViT internals. Abnar and Zuidema [**?**] proposed propagating attention across transformer layers by multiplying per-layer head-averaged attention matrices, giving end-to-end attention flow from any query token to all input tokens. We use rollout to obtain a 576-dimensional [CLS]-to-patch weight vector per image and study its entropy and spatial spread. Raghu et al. [**?**] used linear probing to characterize where object information sits across ViT depth. We apply the same approach with hallucination outcome as the stratifier.

VEGAS and attention-based steering. Seo et al. proposed VEGAS [**?**], which injects ViT [CLS]-to-patch attention into LLM middle layers (typically layers 14–15 of Vicuna-7B) to bias

generation toward visually salient patches. The intervention takes the form of a hidden-state rescaling at image-token positions, blended with uniform attention by a strength parameter λ. The method assumes ViT attention is a reliable steering signal across all input images. Our Track C tests this assumption empirically by stratifying its effect by encoder-quality groups defined either by attention entropy (Track A) or by linear-probe accuracy (Track B).

LLaVA-1.5. Liu et al. [**?**] introduced LLaVA-1.5, which connects a CLIP ViT-L/14@336 visual encoder to a Vicuna-7B language model through a two-layer MLP projector. The model expands the single `<image>` token in the prompt to 576 visual tokens after the projector. All three tracks use `llava-hf/llava-1.5-7b-hf`.

3 Shared Experimental Setup

Model. `llava-hf/llava-1.5-7b-hf` loaded in float16 via `LlavaForConditionalGeneration` (`transformers` 4.37.2). The visual encoder is CLIP ViT-L/14@336 with 24 layers, 16 heads, 1024-dim hidden, and $24 \times 24 = 576$ patch tokens. The language model is Vicuna-7B with 32 decoder layers and a 4096-dim hidden state.

Dataset. 500 randomly sampled images from the COCO 2017 validation set, drawn with seed 639. The same image IDs are used across all three tracks; `sampled_image_ids.json` carries the order.

Hallucination ground truth. CHAIR_i scoring against COCO instance annotations. An image is labelled *hallucinating* if $\text{CHAIR}_i > 0$, that is, the caption mentions at least one object type that is not present in the COCO ground truth for that image. Class balance on our sample: 329 hallucinating, 171 clean (hallucination rate = 65.8%).

Compute. Tracks A and B were run on Google Colab Pro with an A100 (40 GB). Track C v1 was run on a Colab T4 (16 GB) with 4-bit NF4 quantization on the LLM weights (vision tower kept at fp16). Track C v2 was run on an RTX PRO 6000 Blackwell (102 GB) at full fp16, no quantization.

4 Track A: ViT Attention Rollout Analysis

4.1 Question

Does the ViT [CLS] attention concentrate on semantically relevant patches in clean images, and become diffuse or misfocused in hallucinating images?

4.2 Procedure

1. For each of the 500 images, run a forward pass through the LLaVA ViT and capture per-layer attention tensors of shape $(H, 577, 577)$ for all 24 layers, where $H = 16$ heads.
2. Compute attention rollout: head-average each layer to a $(577, 577)$ matrix, add the identity (residual skip), row-normalize, then take the matrix product across all 24 layers.
3. Extract the [CLS] row of the rollout matrix, drop the [CLS] self-entry, and renormalize over

the 576 patch positions. Compute the Shannon entropy of the resulting distribution in nats:

$$H(\text{rollout}) = -\sum_{i=1}^{576} w_i \log w_i$$

4. Compute spatial spread as the normalized variance of patch coordinates weighted by the rollout distribution.
5. Stratify all measures by `is_hallucinating` and split into entropy quartiles (Q_1 = most focused, Q_4 = most diffuse), 125 images per quartile.

The captioning prompt is *"Describe everything you see in this image in detail, listing all objects, people, and animals visible."* which yields long object-enumeration captions.

4.3 Results

Mean CHAIR_i across 500 images is 0.3186 with 329/500 (65.8%) flagged as hallucinating. No statistical test of the attention signal against the hallucination outcome reaches significance (Table 1). The quartile breakdown of hallucination rate is 61.6% → 68.8% → 68.8% → 64.0% from most-focused to most-diffuse quartile. The trend is non-monotonic and the swing is only 2.4 pp. Per-layer entropy curves show the largest hallucinating-vs-clean gap at layer 19 ($\Delta = 0.130$ nats) but this is small relative to within-group variance and runs in the opposite direction at layers 1 through 14.

Table 1: Track A core statistics. Hallucinating images and clean images are statistically indistinguishable on every measure of [CLS] attention geometry we tested.

Test	**Hallucinating** ($n = 329$)	**Clean** ($n = 171$)	**Statistic**	**Verdict**
Mean rollout entropy (nats)	6.3211	6.3207	$d = 0.039$	negligible
Mean spatial spread	0.4277	0.4252	$\Delta = 0.0025$	negligible
Spearman(entropy, CHAIR_i)	—	—	$\rho = 0.003$, $p = 0.94$	null
Spearman(entropy, $\mathbb{1}$[hall])	—	—	$\rho = 0.013$, $p = 0.77$	null
Welch t-test (entropy)	—	—	$t = 0.402$, $p = 0.69$	null
Mann-Whitney U (entropy)	—	—	U stat, $p = 0.39$	null

4.4 Finding

ViT [CLS] attention rollout is statistically indistinguishable between hallucinating and clean images on this 500-image sample. Track A is a clean null. Two things follow.

Rules out. The simple version of our hypothesis where hallucination is driven by spatial misfocus, that is, where the ViT looks at the wrong region of the image and the LLM hallucinates as a consequence.

Does not rule out. An encoder failure that lives in patch features rather than in attention geometry. The information that distinguishes the two subsets may live in what the patches encode, not in which patches are attended to. Track B tests this directly.

5 Track B: Patch Representation Linear Probing

5.1 Question

If attention geometry is uninformative, does the content of patch embeddings encode object identity less reliably on hallucinating images?

5.2 Procedure

1. For each of the 500 images, hook the ViT and capture hidden states at four probe layers $L \in \{6, 12, 18, 24\}$. We changed from the proposed $\{8, 16, 24\}$ to give uniform depth coverage of the 24-layer encoder.
2. For each layer, store both the [CLS] vector (1024-d) and the mean-pooled patch embedding (1024-d).
3. For each of the 80 COCO categories with at least 20 positive examples in our sample, train a binary ℓ_2-regularized logistic regression probe (lbfgs solver) under 5-fold cross-validation, predicting object presence from the mean-pooled patch embedding. 17 categories survive: backpack, bed, bench, book, bottle, bowl, car, cell phone, chair, cup, dining table, handbag, laptop, person, sink, truck, umbrella.
4. For each (layer, category) probe, compute three accuracies and three AUCs: overall, on the hallucinating subset only, and on the clean subset only.
5. Per layer, test $\Delta\text{acc} = \text{acc}_{\text{hall}} - \text{acc}_{\text{clean}}$ across the 17 categories using the Wilcoxon signed-rank test.

A total of $4 \times 17 = 68$ probes were trained.

5.3 Results

Three observations stand out (Table 2). First, mean probe accuracy and AUC rise with depth and peak at layer 18 (acc = 0.946, AUC = 0.905), consistent with the standard finding that mid-late ViT layers carry the most object-discriminative information. Second, probes are uniformly worse on hallucinating images at every layer, with $\Delta\text{acc} < 0$ and $p < 0.005$ in all four cases. Third, the gap is smallest at layer 18 (-0.027) but never closes; the mean across layers is -0.035.

Table 2: Track B linear probe accuracy across ViT depth, stratified by hallucination outcome. All four layers reach significance at $\alpha = 0.005$.

Layer	**Mean Acc**	**Mean AUC**	**Acc_{hall}**	**$\text{Acc}_{\text{clean}}$**	**ΔAcc**	**Wilcoxon p**
6	0.909	0.734	0.895	0.934	-0.039	8.4×10^{-4}
12	0.933	0.833	0.920	0.956	-0.036	6.6×10^{-4}
18	0.946	0.905	0.937	0.964	-0.027	4.2×10^{-3}
24	0.939	0.875	0.926	0.964	-0.038	1.5×10^{-4}
				Mean ΔAcc across all four layers:	-0.035	—

5.4 Finding

Track B confirms an encoder-side signal of hallucination. Mean-pooled patch embeddings on hallucinating images are linearly less informative about object presence than those of clean images at every probed layer. The effect is significant ($p < 0.005$) at all four layers and largest at the input-near and output-near ends of the ViT, with mid-late layers (especially layer 18) partially compensating but not closing the gap.

This is a correlational result. Hallucinating images may be intrinsically harder, with more objects, smaller objects, or rarer categories, and the probe gap may reflect difficulty rather than encoder failure. Track C is the causal test.

5.5 What Tracks A and B Together Force

The combination of Track A null and Track B positive sharpens the hypothesis. Before the experiments we expected encoder failure to show up in two ways, attention and features. The data says it shows up in only one of them.

> **Refined hypothesis.** The ViT encoder fails on hallucinating images, but the failure is in feature content, not attention geometry. Patch embeddings on hallucinating images carry less linearly decodable object information at every probed layer. Spatial attention distributions look essentially identical to clean cases. Methods that trust ViT attention as a corrective signal cannot see this failure mode.

This refined hypothesis directly motivates Track C. The right way to test whether VEGAS, an attention-trusting method, has the predicted blind spot is to stratify its effect by Track B quality groups, not by Track A entropy. We report results for both stratifications below because we ran both.

6 Track C: VEGAS Falsification

6.1 Question

Does VEGAS, an attention-based steering method that injects ViT attention into LLM middle layers, perform differently on encoder-clean versus encoder-degraded images? We address this question through two implementations of VEGAS that differ in quantization, stratification, and the precise steering rule. We report both and discuss the discrepancy.

6.2 Background: the VEGAS Mechanism

VEGAS [**?**] assumes ViT attention is a reliable spatial signal and uses it to steer LLM generation. During decoding, it identifies concentrated visual tokens (high attention weight) and amplifies their influence on LLM middle-layer hidden states, scaled by a visual concentration score α:

$$\alpha = 1 - \frac{H(\text{attention})}{\log N_{\text{patches}}}, \quad \alpha \in [0, 1]$$

where $H(\cdot)$ is Shannon entropy and $N_{\text{patches}} = 576$. High α means the ViT's attention is concentrated; the method then steers more aggressively. Low α means diffuse attention; the method steers weakly. The hook target is LLM decoder layers around 14–15, where the published paper identifies peak conflict between visual and textual signals.

6.3 Track C Version 1: 4-bit Quantized, Entropy-Stratified

Procedure

We loaded LLaVA-1.5-7B at 4-bit NF4 quantization via `bitsandbytes` on a Colab T4. The vision tower was kept at fp16 so the [CLS]-to-patch attention used for steering is numerically exact. We re-generated a vanilla baseline at the same 4-bit precision in the same notebook with the prompt *"Describe this image."* so the VEGAS-vs-baseline comparison is matched on quantization and prompt. The VEGAS hook is a forward pre-hook on LLM decoder layers 14 and 15 that rescales the 576 visual token hidden states by a convex combination of uniform attention and the ViT final-layer [CLS]-to-patch attention with $\lambda = 0.5$, raised to $\lambda = 0.75$ when the normalized [CLS] attention has VABE ≥ 0.5. The raw final-layer attention is sqrt-compressed and clipped to $[0.5, 2.0]$ around its mean to keep the intervention numerically safe.

We stratified images by Track A entropy quartiles (Q_1 = most focused, Q_4 = most diffuse), grouping $Q_1 + Q_2$ as "clean attention" (250 images) and $Q_3 + Q_4$ as "degraded attention" (250 images). Statistical testing used 1000-iteration paired bootstrap to estimate confidence intervals on the per-stratum mean delta.

Results

Table 3 reports per-stratum CHAIR$_i$ for vanilla and VEGAS, the delta, the bootstrap 95% CI, and the bootstrap probability that VEGAS hurts ($\Delta \geq 0$). Two patterns emerge.

Table 3: Track C Version 1 (4-bit NF4, entropy-stratified). VEGAS minus vanilla, per stratum. Negative Δ means VEGAS reduces hallucination.

Stratum	n	**Vanilla CHAIR$_i$**	**VEGAS CHAIR$_i$**	Δ	**95% CI**	$P(\Delta \geq 0)$
Q_1 (low entropy)	125	0.174	0.156	-0.017	$[-0.054, +0.015]$	0.170
Q_2 (mid-low)	125	0.183	0.184	$+0.001$	$[-0.033, +0.029]$	0.519
Q_3 (mid-high)	125	0.227	0.244	$+0.017$	$[-0.022, +0.055]$	0.818
Q_4 (high entropy)	125	0.196	0.221	$+0.025$	$[-0.010, +0.060]$	0.899
Binary clean ($Q_1 + Q_2$)	250	0.179	0.170	-0.008	$[-0.033, +0.015]$	0.266
Binary degraded ($Q_3 + Q_4$)	250	0.211	0.232	$+0.021$	$[-0.009, +0.045]$	**0.929**
All	500	0.195	0.201	$+0.006$	$[-0.012, +0.024]$	0.760

First, the per-quartile delta moves monotonically from -0.017 at Q_1 to $+0.025$ at Q_4. VEGAS helps slightly on the most focused images and hurts progressively as attention becomes more diffuse. Second, the binary split makes the pattern crisp: $\Delta = -0.008$ on the clean half (-1.6 pp hallucination rate) and $\Delta = +0.021$ on the degraded half ($+4.4$ pp). The bootstrap puts $P(\Delta \geq 0) = 0.929$ on the degraded half, directional evidence of VEGAS backfire. The 95% CI does include zero ($[-0.009, +0.045]$), so this is directional rather than definitive at $\alpha = 0.05$. Averaged across all 500 images, the two effects roughly cancel ($\Delta = +0.006$, $P = 0.760$), which would make VEGAS appear neutral overall. The per-stratum breakdown is what reveals the actual signal.

6.4 Track C Version 2: fp16 Full Model, Quality-Group Stratified

Procedure

We re-ran Track C at full fp16 precision on an RTX PRO 6000 Blackwell with 102 GB VRAM, removing quantization entirely. The prompt was *"Describe everything you see in this image in one sentence."* We computed a per-image concentration score from the pre-computed 24-layer ViT rollout:

$$\alpha = 1 - \frac{H(\text{rollout})}{\log 576}, \quad \alpha \in [0, 1]$$

The VEGAS hook fires when $\alpha > 0.01$. When it fires, it identifies the top-20% rollout patches and amplifies their hidden-state contribution at LLM layers 13 to 18 by a factor of $(1 + \alpha)$.

Stratification was by Track B quality group: high-quality (171 images, those classified as clean by Track A's CHAIR labels, mean layer-18 probe accuracy 0.964) versus low-quality (329 images, those classified as hallucinating, mean layer-18 probe accuracy 0.937). This is the stratification that the refined hypothesis from Tracks A+B actually predicts: VEGAS should fail on the encoder-degraded group, not just the diffuse-attention group.

Results

VEGAS produced *zero* change in CHAIR_i on every single image (Table 4). α values were in $[0.002, 0.010]$ with mean ≈ 0.005 across all 500 images. This is consistent with Track A: rollout entropy is near maximum ($\bar{H} \approx 6.32$ nats vs. $\log 576 \approx 6.36$ nats), which makes α near zero by construction. With $\alpha \approx 0.005$, the steering scale $(1 + \alpha) \approx 1.005$ is too small to change which token wins the argmax under greedy decoding. Wilcoxon detects no change. Mann-Whitney U comparing ΔCHAIR_i between groups gives $p = 1.000$. Spearman ρ between α and ΔCHAIR_i is undefined because ΔCHAIR_i is a constant zero vector.

Table 4: Track C Version 2 (fp16, Track B quality-group stratified). VEGAS minus baseline.

Group	n	**Δ Hall. rate**	**Δ CHAIR$_i$**	**Mann-Whitney p**
All images	500	+0.000	+0.0000	—
High-quality (clean)	171	+0.000	+0.0000	—
Low-quality (hallucinating)	329	+0.000	+0.0000	—
High vs. Low (ΔCHAIR_i)	—	—	—	1.000

6.5 Why the Two Versions Disagree

The two implementations share the same intent, namely an attention-guided rescale at LLM middle layers, but they target different signals.

Version 1 uses the raw final-layer ViT [CLS]-to-patch attention as the steering input. Final-layer attention before any rollout is concentrated, with one or two patches typically holding 20–30% of the mass. Version 1 blends this signal with uniform at $\lambda = 0.5$, which gives a meaningful intervention magnitude, and applies it at LLM layers 14–15.

Version 2 computes the concentration score from the pre-computed 24-layer rollout. Rollout is the matrix product of 24 head-averaged attention matrices. Multiplying many near-uniform matrices is

mathematically guaranteed to produce a near-uniform output: the largest concentration any image can achieve is bounded by what the least concentrated layer in the chain provides. The result is that $\alpha = 1 - H/\log 576 \approx 0.005$ for every image, regardless of its actual attention pattern. The threshold $\alpha > 0.01$ blocks most images and the scaling $(1 + \alpha)$ is too small to perturb greedy decoding even on the few that pass.

Both implementations are valid sketches of the published VEGAS idea. Version 1 produces a directional backfire on diffuse-attention images. Version 2 produces no effect because the steering is effectively disabled. The disagreement is not a contradiction; it is the consequence of two design choices for the input signal. Version 1's choice (raw final-layer attention) preserves concentration. Version 2's choice (rollout-derived concentration) does not.

6.6 Combined Finding

The two versions converge on the same higher-level conclusion at different evidentiary strengths. Version 1 provides direct evidence that an attention-based intervention can backfire on diffuse-attention images ($P(\Delta \geq 0) = 0.929$ on the degraded half). Version 2 provides indirect evidence that attention rollout, the natural multi-layer signal for measuring concentration, is too uniform across images to drive a usable intervention at all. Both are consistent with the project's main conclusion: ViT attention is not a reliable signal for selectively correcting hallucination.

7 Discussion

7.1 What the Three Tracks Together Show

Refined hypothesis. Tracks A, B, and C together support a sharper version of the original hypothesis. The ViT encoder contributes measurably to LLaVA-1.5-7B hallucination, but the failure is in feature content (Track B), not in attention geometry (Track A). Patch embeddings on hallucinating images carry less linearly decodable object information at every probed layer, while spatial attention distributions look essentially identical to clean cases. Attention-based steering methods that trust ViT attention as a corrective signal cannot address this failure mode because their corrective signal is computed from the very attention quantity Track A showed to be uninformative.

The Track C structural blind spot. VEGAS's core assumption is that ViT attention concentration identifies trustworthy visual tokens. Our three-track investigation challenges this assumption on two levels. Track A nullifies α as a discriminator: the concentration score (whether computed from raw attention or rollout) does not differ between hallucinating and clean images in a way that lets the method detect which images have unreliable encodings. Track B identifies the actual failure, namely degraded patch features, which α cannot see. Track C then confirms the consequence: in version 1 the method backfires when given diffuse attention as input, and in version 2 the method cannot activate at all because α is universally near zero.

7.2 What Each Track Contributes

7.3 Limitations

Sample size. 500 images from a single split of COCO 2017. Track B effect sizes ($\Delta\text{acc} \approx -0.035$) are statistically significant but modest. The Track C v1 binary-degraded result is directionally strong ($P = 0.929$) but its 95% CI includes zero ($[-0.009, +0.045]$). Replication on a larger or different sample would tighten the inference.

Table 5: Summary of contributions across the three tracks.

Track	Question	Key result	Implication
A: Attention rollout	Is hallucination predicted by diffuse ViT attention?	$\rho = 0.003$, $p = 0.94$, $d = 0.039$. Complete null.	Rules out attention misfocus as the failure mode.
B: Linear probing	Is hallucination predicted by degraded patch features?	$\Delta\text{acc} \in [-0.039, -0.027]$, $p < 0.005$ at all 4 layers.	Feature degradation is real; the encoder is a measurable contributor.
C: VEGAS (v1)	Does VEGAS fail on diffuse-attention images?	$\Delta = +0.021$ on degraded half, $P(\Delta \geq 0) = 0.929$. Directional backfire.	Attention-based steering is unreliable in the diffuse regime.
C: VEGAS (v2)	Does VEGAS fail on feature-degraded images?	$\Delta = 0.0000$ all groups; $\alpha \approx 0$ for all images.	Rollout-based α cannot drive steering at all.

Causality. Track B is correlational. The same images that yield low probe accuracy might be intrinsically harder (more objects, smaller objects, rarer categories), and the gap may reflect difficulty rather than encoder failure. Track C was designed to convert correlation to causation but neither version landed a causal claim. Version 1 stratifies by entropy, which Track A showed does not predict hallucination. Version 2 stratifies by Track B quality groups but the steering was effectively disabled by the rollout-derived α.

Track C implementations. The two versions of Track C use different signals for the steering input (raw final-layer attention vs. rollout-derived α), different prompts, and different stratifications. We report both because we ran both, but the disagreement between them is itself a limitation: it shows that the result depends on implementation choices that the published VEGAS paper does not fully specify.

Scope cuts. Of the four experiments listed in the original proposal, the VMamba and ImageNet-C robustness comparison was not run due to compute and time constraints. POPE evaluation across three adversarial splits was scoped out in favour of CHAIR_i as the primary metric. Both are recorded in Future Work.

Probe categories. Only 17 of 80 COCO categories had at least 20 positive examples in the 500-image sample. Rare categories may behave differently and were not included in the probe analysis.

8 Conclusions and Future Work

8.1 Conclusions

We ran three controlled experiments on whether LLaVA-1.5-7B's ViT visual encoder is failing on images that lead to hallucination. The short answer is yes, but not in the way the literature usually expects. Track A finds no signal in attention geometry. Track B finds a consistent significant signal in patch-embedding feature content. Track C, in two configurations, shows that an attention-based mitigation method either backfires on diffuse-attention images (4-bit, entropy-stratified) or fails to activate at all because rollout attention is uniformly diffuse (fp16, quality-group-stratified).

The three numbers that carry the project: $\rho = 0.003$ (Track A null), $\Delta\text{acc} = -0.035$ across layers at

$p < 0.005$ (Track B positive), and $P(\Delta \geq 0) = 0.929$ on the degraded half of Track C v1 (directional backfire). Methods that trust ViT attention as a corrective signal address the wrong quantity. Future hallucination mitigation for LLaVA-style models should target patch embedding quality at mid-late ViT layers (around layer 18) rather than attention redistribution.

8.2 Future Work

Reconcile the two Track C versions. The most useful next step is a third Track C run that combines the strengths of both versions: full fp16 model (no quantization confound), raw final-layer [CLS]-to-patch attention as the steering signal (so the intervention has real magnitude), blending at $\lambda = 0.5$ as in version 1, and stratifying by Track B per-image probe accuracy (since probe accuracy is correlated with hallucination outcome and entropy is not). This requires Track B to export per-image probe accuracy, which the current outputs do not contain. Generating that file is straightforward from the cached patch embeddings and the trained probes; it does not require any new model inference. The resulting experiment would be the cleanest causal test of whether VEGAS specifically fails on encoder-degraded images.

Investigating the discrepancy between Track C versions. The two versions differ in three axes (precision, prompt, steering signal) and in the stratifier. A factorial follow-up could vary each axis independently: (i) full fp16 with raw final-layer attention (predicted: similar to v1 directionality but tighter intervals); (ii) 4-bit with rollout-derived α (predicted: zero effect, like v2, confirming the signal choice is the cause); (iii) cross-stratifier robustness, running v1's intervention with both entropy and probe-accuracy stratifications and asking whether the backfire is stronger on the latter. The hypothesis these experiments test is that the Track C result is determined almost entirely by the steering signal choice, with quantization, prompt, and stratifier as second-order factors.

Causal patch ablation. Zero-ablate the lowest-probe-accuracy patches at layer 18 and measure CHAIR_i change. Required controls: random-patch ablation (matched count, drawn uniformly), sham-layer ablation (same procedure at layer 6 where probe accuracy is lower), and reversal test (ablate high-quality patches in clean images). A targeted ablation that produces a significantly larger CHAIR_i change than these controls converts Track B's correlation into a causal claim. Pre-registered effect size: paired Wilcoxon, minimum detectable $\Delta\text{CHAIR}_i = 0.05$, $\alpha = 0.01$, two-sided.

Counterfactual patch swap. For an image A where LLaVA hallucinates object X, find a clean image B that does not contain X and swap the patches of B corresponding to X's expected location into A. If the hallucination disappears, the hallucination travels with patch features rather than with global context. This is the strongest possible evidence that the failure is encoded in patch content.

VMamba and ImageNet-C robustness. The fourth experiment in the original proposal was not run. Compare ViT-L/14 against VMamba-Small on a 5000-image stratified subsample of ImageNet-C across 15 corruption types and 5 severity levels. The hypothesis is that fine-grained corruptions (pixel noise, pixelation, JPEG) trigger the patch-spread failure mode our Track B results imply, while coarse corruptions (blur, brightness) do not. SSM-based encoders, by their sequential design, should not exhibit this asymmetry.

POPE evaluation. Run POPE across the three adversarial splits (random, popular, adversarial) on the existing 500 captions. POPE asks binary questions and does not require new inference; it can be applied to the saved captions directly. Adding it would let us check whether the Track A null and Track B positive replicate under a binary-probe metric.

Code and data. Notebooks, intermediate CSVs, figures, and the report sources are available at https://github.com/NikhilAdyapak/vit-hallucination-llava.

References

[1] Li, Y., Du, Y., Zhou, K., Wang, J., Zhao, W. X., and Wen, J.-R. Evaluating object hallucination in large vision-language models. *EMNLP*, 2023.

[2] Rohrbach, A., Hendricks, L. A., Burns, K., Darrell, T., and Saenko, K. Object hallucination in image captioning. *EMNLP*, 2018.

[3] Abnar, S., and Zuidema, W. Quantifying attention flow in transformers. *ACL*, 2020. arXiv:2005.00928.

[4] Seo, J., et al. VEGAS: ViT-attention-guided steering for VLM hallucination mitigation. arXiv:2512.12089, 2024.

[5] Raghu, M., Unterthiner, T., Kornblith, S., Zhang, C., and Dosovitskiy, A. Do vision transformers see like convolutional neural networks? *NeurIPS*, 2021.

[6] Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. *NeurIPS*, 2023.

[7] Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al. An image is worth 16×16 words: Transformers for image recognition at scale. *ICLR*, 2021. arXiv:2010.11929.

[8] Seo, J., et al. ViT hallucination analysis in vision-language models. arXiv:2510.09008, 2025.

A Track A: Per-Layer Entropy Gap

Table 6: Per-layer entropy gap (hallucinating − clean) across all 24 ViT layers.

Layer range	Gap Δ (nats)	Direction
Layers 1–13	≈ 0.01 to 0.02	hallucinating > clean (small)
Layer 14	≈ 0.00 (crossing)	—
Layers 15–24	rises to 0.130 at layer 19	hallucinating > clean

The gap is largest at layer 19 ($\Delta = 0.130$ nats) but small relative to within-group variance ($\sigma_{\text{within}} \approx 0.4$ nats). It does not translate into a predictive signal at the image level, as Spearman, Welch t, and Mann-Whitney all return null on the rollout-level entropy.

B Track B: Probe Categories

The 17 COCO categories with at least 20 positive examples in the 500-image sample, used for the linear probe analysis: *backpack, bed, bench, book, bottle, bowl, car, cell phone, chair, cup, dining table, handbag, laptop, person, sink, truck, umbrella.*

C Track C: Observed Concentration Distribution (Version 2)

All 500 images had concentration $\alpha \in [0.002, 0.010]$ with mean ≈ 0.005. This is consistent with the Track A finding that rollout entropy is nearly maximal ($\bar{H} \approx 6.321$ nats, max $= \log 576 \approx 6.356$ nats). The gap between observed mean entropy and maximum entropy is only ≈ 0.035 nats, which maps to $\alpha \approx 0.005$. VEGAS steers by $(1 + \alpha) \approx 1.005$, effectively unity, which does not change greedy-decoded outputs on any image.

D Repository Layout

The accompanying repository (https://github.com/NikhilAdyapak/vit-hallucination-llava) contains:

- `notebooks/track_a/` — LLaVA inference + CHAIR scoring (notebook 1) and ViT attention rollout extraction (notebook 2).
- `notebooks/track_c/` — three Track C variants: `v0_initial_attempt.ipynb` (superseded, fp16 vs. 4-bit comparison issue), `v1_4bit_entropy_stratified.ipynb` (Version 1 in this paper), `v2_fp16_quality_stratified.ipynb` (Version 2 in this paper). The `trackc_package/` directory contains the Python module used by Version 1's ablation pipeline.
- `data/track_a_outputs/` — `chair_labels.csv`, `entropy_ranks.csv`, `sampled_image_ids.json`.
- `data/track_c_outputs/` — vanilla and VEGAS captions (4-bit) and the bootstrap delta tables for Version 1.
- `figures/` — Track A entropy distribution, per-layer entropy, quartile analysis, Spearman scatter.
- `reports/` — this final report, the proposal, and the Track A + B progress report.